

AL-JAWF KUFRA, Libyan Arab Jamahiriya

WMO: 622710

Lat: 24.1986N

Lon: 23.3000E

Elev: 436

StdP: 96.20

Time Zone: 2.00 (E02)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.2	7.4	-12.0	1.4	19.1	-10.6	1.6	20.1	9.7	16.1	8.8	16.6	1.9	50	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	42.0	18.6	40.6	18.4	39.6	18.2	19.9	38.1	19.3	37.7	18.8	37.3	3.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	9.3	28.3	11.0	8.6	28.4	10.1	8.1	28.9	58.4	38.0	56.3	37.6	54.7	37.4	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.1	7.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.9	14.2	16.9	21.1	26.7	29.9	32.4	32.6	32.2	30.3	26.3	20.8	15.5
	DBStd	7.22	2.95	3.48	3.99	3.34	3.36	2.32	1.58	1.47	2.29	3.00	2.95	2.60
	HDD10.0	1	1	1	0	0	0	0	0	0	0	0	0	0
	HDD18.3	320	133	64	16	0	0	0	0	0	0	10	97	
	CDD10.0	5453	132	192	344	499	616	672	700	689	608	504	324	171
	CDD18.3	2730	6	23	102	249	358	422	442	431	358	246	84	9
	CDH23.3	39022	72	294	1124	3077	5190	6753	7104	6737	5131	2741	712	87
	CDH26.7	23588	10	108	520	1672	3218	4476	4654	4296	3038	1348	233	17
Wind	WSAvg	3.4	2.8	3.0	3.6	4.2	3.6	3.7	3.8	3.8	3.5	3.4	2.9	2.7
Precipitation	PrecAvg	2	1	0	0	0	0	0	0	0	0	0	0	0
	PrecMax	18	10	7	4	8	3	0	1	0	0	2	8	8
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	4	2	1	1	1	0	0	0	0	0	0	1	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	33.7	37.9	40.6	43.9	45.0	42.1	40.8	40.6	38.9	33.5	28.5
		MCWB	12.4	14.0	16.3	17.4	18.7	19.3	19.0	19.0	19.2	17.9	15.1	13.0
	2%	DB	25.3	30.0	34.8	38.4	41.9	42.4	40.8	39.7	39.5	36.6	31.3	25.8
		MCWB	11.2	12.3	14.8	16.5	18.3	18.6	19.0	18.7	18.6	17.5	14.6	12.1
	5%	DB	23.3	26.9	32.2	36.6	40.0	40.5	39.3	38.7	38.1	34.7	29.4	23.2
		MCWB	10.5	11.8	13.7	15.8	17.7	18.1	18.6	18.4	18.2	17.0	14.3	11.3
	10%	DB	21.0	24.2	29.5	34.5	37.9	39.1	38.2	37.7	36.6	33.0	27.6	21.1
		MCWB	9.7	10.6	12.7	15.2	17.1	17.8	18.3	18.1	17.8	16.4	13.8	10.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	15.3	16.8	18.3	19.2	19.9	21.4	20.5	20.3	19.3	17.0	13.8
		MCDB	25.8	31.7	36.5	36.8	39.9	41.7	38.4	38.0	37.8	32.9	28.1	25.6
	2%	WB	11.8	13.2	15.0	17.1	18.6	19.2	20.1	19.6	19.3	18.2	15.8	12.7
		MCDB	23.8	27.5	33.4	36.7	39.9	40.5	37.9	37.2	37.0	33.7	27.7	23.4
	5%	WB	11.0	11.9	14.0	16.2	18.0	18.5	19.2	19.0	18.6	17.6	15.1	11.9
		MCDB	21.0	25.8	31.5	34.9	39.6	39.4	37.3	36.8	36.1	33.1	27.4	21.7
	10%	WB	10.2	11.0	13.0	15.4	17.2	18.0	18.7	18.5	18.1	16.9	14.3	11.2
		MCDB	19.6	22.8	28.8	33.6	36.9	38.4	36.8	36.4	35.3	31.7	26.2	20.1
Mean Daily Temperature Range	5% DB	MDBR	11.7	13.0	13.1	13.4	13.0	12.8	11.3	10.9	11.5	11.6	11.9	11.2
		MCDBR	14.7	17.5	16.9	15.7	15.2	14.4	12.3	11.6	13.1	13.2	13.9	13.8
		MCWBR	6.8	7.9	7.4	6.3	6.0	5.4	4.4	4.0	4.3	4.4	5.7	6.0
	5% WB	MCDBR	13.2	15.8	15.6	14.3	14.2	13.5	11.4	10.9	11.9	12.1	12.4	12.1
		MCWBR	6.2	7.3	7.0	5.9	5.7	5.0	4.3	4.0	4.2	4.3	5.2	5.4
Clear-Sky Solar Irradiance	taub	0.362	0.414	0.473	0.593	0.613	0.531	0.503	0.496	0.482	0.440	0.376	0.362	
	taud	2.112	1.934	1.784	1.560	1.540	1.722	1.797	1.821	1.849	1.934	2.088	2.109	
	Ebn at Noon	885	859	827	740	722	778	800	806	808	822	861	867	
	Edh at Noon	140	180	220	281	286	236	219	214	203	177	142	136	
All-Sky Solar Radiation	RadAvg	4.78	5.58	6.61	7.32	7.71	8.02	7.95	7.58	6.89	5.99	5.01	4.48	
	RadStd	0.14	0.25	0.29	0.20	0.22	0.19	0.10	0.12	0.11	0.12	0.10	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page